

Do ESG Scores Predict Stock Returns?

Evidence from Indian Large-Cap Companies (2020–2025)

Abstract

The relationship between corporate ESG (Environmental, Social and Governance) performance and financial returns has emerged as one of the most actively contested questions in sustainable finance. While European equity markets have documented a meaningful ESG premium structurally reinforced by the EU Sustainable Finance Disclosure Regulation (SFDR) and the EU Taxonomy, which collectively channel approximately EUR 3.9 trillion in assets toward sustainability-classified funds [3][4] evidence for emerging markets such as India remains considerably sparser and more mixed [1][12]. This paper investigates whether ESG scores predict stock returns for 100 NIFTY large-cap companies over 2020–2025. Splitting the sample at the median ESG score of 63, High-ESG companies (n=51, mean score 68.3) outperformed Low-ESG companies (n=49, mean score 56.3) across all four holding periods examined, with the widest gap at the three-year horizon (+3.24 percentage points). However, the Pearson correlation between ESG score and five-year CAGR is weak and not statistically significant at the five percent level ($r = 0.179$, $p = 0.075$, $R^2 = 0.032$), indicating that ESG scores alone explain very little of the variation in individual company returns. Sector effects are substantial: Low-ESG companies lead in four of eleven comparable sectors, including Pharmaceuticals and IT. The findings are broadly consistent with emerging-market literature [1][12][13] and contrast with stronger premia documented in Europe where regulatory capital flows mechanically reprice high-ESG securities [3][4][5]. The ESG premium in India appears nascent, and this paper provides a quantitative baseline for tracking its evolution as the SEBI BRSR framework matures [15].

Keywords: *ESG scores, stock returns, India, large-cap companies, Pearson correlation, NIFTY, sustainable finance, SEBI BRS*

1. Introduction

The global sustainable-finance landscape has undergone a profound transformation over the past decade. ESG-labelled assets under management surpassed USD 30 trillion by 2023 [17], and the proliferation of ESG rating providers, green indices and sustainability-linked investment products has made ESG scores a central variable in capital allocation decisions. Meta-analyses synthesising thousands of individual studies find that the majority of empirical evidence points to a non-negative relationship between ESG criteria and corporate financial performance [5][13], lending broad support to the thesis that sustainability and profitability are complementary rather than conflicting objectives. In Europe, this relationship has been materially reinforced by a

regulatory architecture that creates structural demand for high-ESG securities. The EU Sustainable Finance Disclosure Regulation [3] requires asset managers to classify funds as Article 6, Article 8 or Article 9 on the basis of their ESG integration. By 2024, Article 8 and Article 9 funds collectively managed approximately EUR 3.9 trillion [4], mechanically channelling institutional capital toward companies rated favourably by major ESG providers such as MSCI and Sustainalytics [7]. The EU Taxonomy [3] further reinforces this by defining which economic activities qualify as environmentally sustainable, guiding the allocation of credit and investment across the block. This configuration which may be termed the *regulatory channel* creates upward price pressure on high-ESG stocks in European markets, amplifying a premium that may not be fully explained by fundamental improvements in firm performance alone. India presents a markedly different institutional environment. The Securities and Exchange Board of India (SEBI) introduced the Business Responsibility and Sustainability Report (BRSR) framework as a mandatory disclosure requirement for the top 1,000 listed entities from financial year 2022–23 [15], establishing a standardised basis for ESG reporting that did not previously exist. However, the downstream infrastructure ESG-screened fund classifications, a formal green taxonomy, large-scale institutional adoption of ESG investment mandates remains at an early stage relative to European benchmarks. ESG-themed mutual funds in India account for a small fraction of total equity AUM, and the regulatory channel through which ESG scores are translated into equity prices operates at considerably reduced intensity. Against this backdrop, this paper investigates three questions: (1) **Do High-ESG Indian large-cap companies deliver superior average stock returns relative to Low-ESG peers across multiple holding periods?** (2) **Is there a statistically significant linear correlation between ESG scores and Compound Annual Growth Rates at the individual company level?** (3) **Are sector effects sufficiently large to moderate, or even reverse, the aggregate ESG–return relationship?** The study contributes to the nascent empirical literature on ESG investing in Indian markets [1] and complements earlier work by the author on sustainable finance access for Indian MSMEs [10], extending the research programme from the financing side of sustainable development to the investment–return dimension.

2. Literature Review and European Context

2.1 Global evidence on ESG and financial performance

The most comprehensive survey of the global ESG–financial performance literature was conducted by Friede, Busch and Bassen [5], who aggregated evidence from more than 2,200 individual empirical studies. Their analysis found that approximately 90 percent of studies documented a non-negative relationship between ESG criteria and corporate financial performance, with the majority reporting a positive effect. Revelli and Viviani [13] reached similar conclusions in a meta-analysis of socially responsible investing (SRI) research, finding no systematic evidence that SRI funds underperform conventional benchmarks on a risk-adjusted basis and some evidence of modest outperformance in certain geographies and time periods. At the firm level, Khan, Serafeim and Yoon [6] provided influential evidence that the ESG–return relationship is modulated by *materiality*: companies performing well on ESG dimensions that are financially material to their specific industry earn significantly higher risk-adjusted returns than peers performing well on immaterial dimensions. This finding implies that aggregate ESG scores

which weight environmental, social and governance performance equally regardless of sector relevance may be noisier predictors of returns than materiality-weighted measures. Pedersen, Fitzgibbons and Pomorski [11] formalised the conditions under which an ESG premium can persist in equilibrium: when a sufficient share of market participants has a preference for high-ESG assets whether ethical, regulatory or reputational a price premium emerges that can be sustained even absent direct ESG effects on fundamentals.

2.2 The European ESG premium and the regulatory channel

European equity markets have been a particularly important laboratory for testing the ESG return relationship in the context of active regulatory intervention. Bauer, Koedijk and Otten [2] provided early evidence that European ethical funds did not systematically underperform conventional peers, with some evidence of outperformance after controlling for style exposures. More recent research documents that the ESG premium in European equities has intensified following the introduction of the SFDR in 2021 [3], with Article 8 and Article 9 fund inflows creating measurable demand-side price effects for high-ESG rated securities [4]. Collectively, these developments establish a regulatory demand channel for ESG that is largely absent in emerging markets at comparable scale. The EU Taxonomy [3] and the Corporate Sustainability Reporting Directive (CSRD) further institutionalise ESG within the European financial system by requiring banks, insurers and asset managers to disclose the sustainability of their assets against a detailed technical classification. These cascading requirements have elevated ESG scores from niche inputs for impact investors to mainstream screening criteria for a large segment of European institutional capital. The UNPRI [17] estimates that signatory assets under management exceed USD 100 trillion globally, though the depth of ESG integration varies substantially across signatories and jurisdictions with Europe generally showing deeper implementation than Asia or emerging markets.

2.3 ESG and returns in emerging markets

Evidence for the ESG–return relationship in emerging markets is considerably sparser and more heterogeneous than in developed economies. Renneboog, Ter Horst and Zhang [12] documented in a cross-country study of ethical funds that financial performance varies markedly by region, with emerging markets showing weaker and less consistent evidence of ESG-linked outperformance. Several structural factors have been advanced to explain this divergence: shallower capital markets with higher information asymmetries make it harder for ESG signals to be efficiently priced; concentrated ownership structures including high promoter holdings, common in India limit the ability of minority institutional investors to reward or penalise ESG performance through capital allocation; and nascent ESG-linked fund flows mean that the demand premium for high-ESG securities is structurally constrained.

Within India specifically, Agrawal [1] examined the ESG–return relationship among large-cap Indian equities and found weak evidence for an ESG premium, attributing this to the early stage of ESG disclosure standardisation and limited institutional adoption of ESG investment strategies at the time of the study. The introduction of SEBI's BRSR mandate [15] and the establishment of the NSE Sustainability Ratings and Analytics Limited (NSRAL) [9] as a dedicated ESG rating subsidiary of the NSE represent material improvements in India's ESG disclosure infrastructure since that study, warranting an updated quantitative examination which

the present paper provides.

3. Data and Methodology

3.1 Sample construction and study period

The sample comprises 100 NIFTY large-cap companies drawn from the NSE large-cap universe [8], selected on the basis of complete data availability across all required variables. The companies span 17 sectors including Banks, Information Technology, Pharmaceuticals, Consumer Goods, Automobiles, Power, Metals & Mining, Cement, Telecom, Industrial Manufacturing, Construction, Finance (Non-Banking), Oil & Gas, Chemicals, Fertilisers & Pesticides, Textiles and Services providing broad representation of the Indian large-cap equity market. The study period is 2020–2025, with stock return data captured as Compound Annual Growth Rates (CAGR) measured as of April–May 2026 across four standard windows: 1-year, 3-year, 5-year and 10-year. The 5-year CAGR serves as the primary return variable as it most closely **approximates a 2020–2025 holding period**.

3.2 Data sources

ESG scores and grade. ESG scores (on a 0–100 scale) and letter grades (A+, A, B+, B, C+) were sourced and cross-referenced from four publicly available disclosure channels: (i) Business Responsibility and Sustainability Reports (BRSR) filed under SEBI Circular SEBI/LAP/GEN/2021/0000000009 [15], accessible on company investor relations pages and the BSE/NSE filings portals; (ii) NSE Sustainability Ratings and Analytics Limited (NSRAL) company-specific ESG rating disclosures for FY 2024–25 [9]; (iii) CRISIL ESG assessment summaries where publicly available; and (iv) the Stakeholders Empowerment Services (SES) ESG ratings report [16]. (v) S&P Global. Where discrepancies existed across sources, scores were manually verified against primary BRSR disclosures and reconciled accordingly.

Stock return data. All CAGR figures across 1-year, 3-year, 5-year and 10-year windows were obtained from Screener.in [14], a publicly accessible Indian equity research platform that aggregates price-adjusted return data from NSE and BSE exchange feeds. CAGRs are computed from historical adjusted closing prices, accounting for corporate actions including stock splits and bonus issues.

Financial control variables. Market capitalisation (₹ crore), Price-to-Earnings (P/E) ratio, 3-year average Return on Equity (ROE, %), Debt-to-Equity ratio, Promoter Holding (%) and Foreign Institutional Investor (FII) Holding (%) were sourced from Screener.in [14], which aggregates data from quarterly BSE/NSE shareholding pattern disclosures and company financial filings submitted to the exchanges under SEBI's continuous disclosure requirements [15].

3.3 Analytical methodology

Companies were classified into High-ESG and Low-ESG groups based on whether their ESG score was at or above the sample median of 63, yielding 51 High-ESG companies (mean score 68.29) and 49 Low-ESG companies (mean score 56.33). Group-level average CAGRs were

computed for all four holding periods. Pearson correlation coefficients between ESG score and each CAGR variable were calculated using the `scipy.stats.pearsonr` function in Python 3.11, with two-tailed p-values reported for the null hypothesis of zero correlation. Sector-level analysis compared mean 5-year CAGR between the High- and Low-ESG subgroups within each of the 11 sectors for which at least one company from each group was available. All descriptive statistics were computed using the `pandas` library in Python 3.11.

4. Findings

4.1 Descriptive statistics

Table 1 presents descriptive statistics for the full sample. ESG scores range from 42 to 81 with a mean of 62.43 and standard deviation of 7.68, reflecting moderate dispersion across the 0–100 scale. The grade distribution is: A+ (7 companies, 7%), A (9, 9%), B+ (21, 21%), B (29, 29%) and C+ (34, 34%), with the bulk of the sample concentrated in the two lower-tier categories. This distribution reflects the current state of ESG performance among Indian large-caps relatively few top-tier performers and a large mid-to-lower tier and is consistent with the assessment of NSRAL [9] and other Indian ESG rating agencies that widespread adoption of best-practice ESG governance remains work in progress. Five-year CAGR ranges from –18% to 65% with a mean of 12.40%, reflecting the substantial dispersion in individual company performance over the period.

Table 1. Descriptive Statistics Full Sample (n = 100, NIFTY Large-Cap, 2020–2025)

Variable	n	Mean	Std Dev	Min	Max
ESG Score (0–100)	100	62.43	7.68	42	81
1-Year CAGR (%)	100	13.30	23.06	–45.00	110.00
3-Year CAGR (%)	100	11.04	13.60	–25.00	85.00
5-Year CAGR (%) [primary]	100	12.40	11.46	–18.00	65.00
10-Year CAGR (%)	100	13.53	7.95	–18.00	45.00
Market Cap (₹ Crore)	100	2,18,871	2,92,153	4,747	19,10,389
P/E Ratio	100	39.18	24.44	8.46	145.00
3-Year Avg ROE (%)	100	21.57	21.28	2.40	157.48

Note. Sources: ESG scores from BRSR disclosures [15], NSRAL [9], CRISIL and SES [16]; CAGR and financial data from Screener.in [14]. Market Cap in ₹ crore. CAGR measured as of April–May 2026.

4.2 High-ESG versus Low-ESG group comparison

Table 2 and Figure 1 present the core group-level comparison. High-ESG companies outperformed Low-ESG companies on average across all four holding periods. The performance

gap is widest at the three-year horizon (+3.24 percentage points: High-ESG 12.63% vs Low-ESG 9.39%), followed by the five-year horizon (+2.87 pp: 13.80% vs 10.94%), the one-year horizon (+1.31 pp: 13.94% vs 12.63%) and the ten-year horizon (+0.32 pp: 13.69% vs 13.37%). The consistent directional outperformance across all four timeframes is noteworthy and provides aggregate-level support for a positive ESG–return association. However, two confounding factors warrant attention. First, High-ESG companies have a substantially higher average market capitalisation (₹2,70,480 crore vs ₹1,65,156 crore for Low-ESG), and there is extensive evidence in the asset pricing literature that size is a significant predictor of return patterns. Second, High-ESG companies exhibit materially higher average three-year Return on Equity (26.69% vs 16.25%), suggesting that the return gap may partially reflect fundamental differences in profitability rather than ESG-specific investor repricing [6][11].

Table 2. High-ESG vs Low-ESG Group Comparison — Average Returns and Financial Characteristics

Metric	High ESG (n=51)	Low ESG (n=49)	Difference (H–L)	Direction
Avg ESG Score	68.29	56.33	+11.96	—
1-Year CAGR (%)	13.94	12.63	+1.31	High ESG ↑
3-Year CAGR (%)	12.63	9.39	+3.24	High ESG ↑
5-Year CAGR (%) ★	13.80	10.94	+2.87	High ESG ↑
10-Year CAGR (%)	13.69	13.37	+0.32	High ESG ↑
Avg Market Cap (₹ Cr)	2,70,480	1,65,156	+1,05,324	—
Avg 3-Year ROE (%)	26.69	16.25	+10.44	—

Note. ★ Primary return variable. Groups defined by ESG score ≥ 63 (High) or < 63 (Low). CAGR in %; Market Cap in ₹ crore. Sources: Screener.in [14]; ESG scores from BRSR [15], NSRAL [9], CRISIL and SES [16].

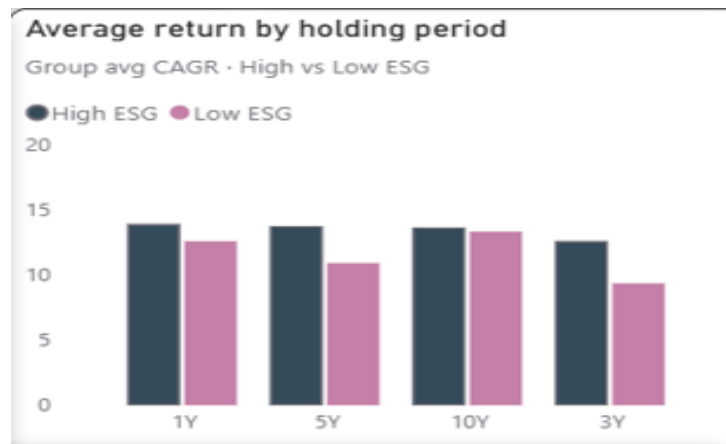


Figure 1. Average stock CAGR (%) by ESG group across four holding periods. High-ESG companies outperform across all timeframes; the gap is widest at the 3-year horizon (+3.24 pp, High-ESG 12.63% vs Low-ESG 9.39%). Source: Author analysis; data from Screener.in [14] and ESG disclosures [9][15][16].

4.3 Pearson correlation analysis

Table 3 and Figure 2 present the correlation results. The Pearson correlation coefficient between ESG score and 5-year CAGR is $r = 0.179$ ($p = 0.075$), indicating a weak positive relationship that falls just short of significance at the conventional five percent threshold. The coefficient of determination ($R^2 = 0.032$) implies that ESG scores explain only approximately 3.2 percent of the variation in five-year stock returns across the sample a low figure that cautions against attributing the group-level outperformance documented in Section 4.2 primarily to a strong or systematic linear ESG–return relationship at the individual company level.

Results are broadly consistent across holding periods (Table 3), with the correlation near-zero at one year ($r = 0.006$, $p = 0.954$) and weakly positive but statistically insignificant at three, five and ten years. These findings align with the pattern documented in the emerging-market literature [1][12] and are consistent with the interpretation that, in the absence of the structural ESG demand channel operating in European markets [3][4][5], ESG scores in India are not yet systematically repriced into equity valuations in the way that asset pricing theory predicts should occur when a large fraction of market participants allocates capital on ESG criteria [11].

Table 3. Pearson Correlation — ESG Score vs Stock CAGR Across All Holding Periods (n = 100)

Return Metric	Pearson r	p-value	Interpretation
1-Year CAGR (%)	0.006	0.954	Negligible — no linear relationship
3-Year CAGR (%)	0.154	0.127	Weak positive — not significant at 5%
5-Year CAGR (%) ★	0.179	0.075†	Weak positive — significant at 10% only
10-Year CAGR (%)	0.132	0.191	Weak positive — not significant at 5%

Note. † Significant at the 10% level ($p < 0.10$) but not at the 5% level. Two-tailed p -values computed using `scipy.stats.pearsonr` (Python 3.11). ★ Primary return variable.



Figure 2. Scatter plot of ESG score against 5-year stock CAGR for 100 NIFTY large-cap companies. The near-flat trendline ($r = 0.179$, $p = 0.075$, $R^2 = 0.032$) illustrates the weak linear association between ESG score and individual company returns over 2020–2025.

4.4 Sector-level analysis

Table 4 and Figure 3 present the sector-level breakdown of 5-year CAGR for the 11 sectors for which comparable High- and Low-ESG data were available. High-ESG companies outperformed in 7 sectors (Automobile, Banks, Cement, Consumer Goods, Metals & Mining, Power and Telecom) and Low-ESG companies outperformed in 4 sectors (Construction, Finance Non-Banking, IT and Pharma). Six sectors Chemicals, Fertilisers & Pesticides, Industrial Manufacturing, Oil & Gas, Services and Textiles had insufficient representation in one ESG group to permit meaningful comparison.

The heterogeneity of sector-level outcomes is a critical finding. The Pharmaceuticals sector presents a clear counter-example to the aggregate ESG premium: Low-ESG Pharma companies averaged 11.0% 5-year CAGR versus 6.8% for High-ESG peers a 4.2 pp differential in the opposite direction to the overall sample finding. Similarly, the IT sector shows Low-ESG outperformance (19.0% vs 10.1%), though the Low-ESG IT sample comprises only one company, limiting the generalisability of this specific comparison. These findings are broadly consistent with the materiality framework advanced by Khan, Serafeim and Yoon [6], which predicts that ESG performance on factors not material to a sector’s value creation model will not systematically predict returns. The Power sector illustrates an outlier concern in the opposite direction: the single High-ESG Power company (Adani Green Energy Ltd., ESG score 66, 5-year CAGR 65%) drives the sector average substantially, a pattern that reflects idiosyncratic firm performance rather than a sector-wide ESG premium.

Table 4. Sector-Wise 5-Year CAGR Comparison: High-ESG vs Low-ESG Companies (2020–2025)

Sector	High ESG CAGR (%)	n	Low ESG CAGR (%)	n	Leader
Automobile	16.4	5	8.8	4	High ESG
Banks	10.4	5	−5.3	3	High ESG
Cement	15.5	2	10.7	3	High ESG
Construction	15.0	1	28.0	1	Low ESG
Consumer Goods	12.2	14	11.4	9	High ESG
Finance: Non-Banking	8.2	4	10.4	5	Low ESG
IT	10.1	7	19.0	1	Low ESG
Metals & Mining	15.2	4	11.5	2	High ESG
Pharma	6.8	4	11.0	11	Low ESG
Power	65.0†	1	35.0	1	High ESG
Telecom	25.0	1	−2.0	1	High ESG

Note. † Power sector High-ESG figure driven primarily by a single company (Adani Green Energy Ltd., 5Y CAGR 65%). Sectors with n = 0 in either group excluded. Sources: Screener.in [14]; ESG scores from BRSR [15], NSRAL [9], CRISIL and SES [16].

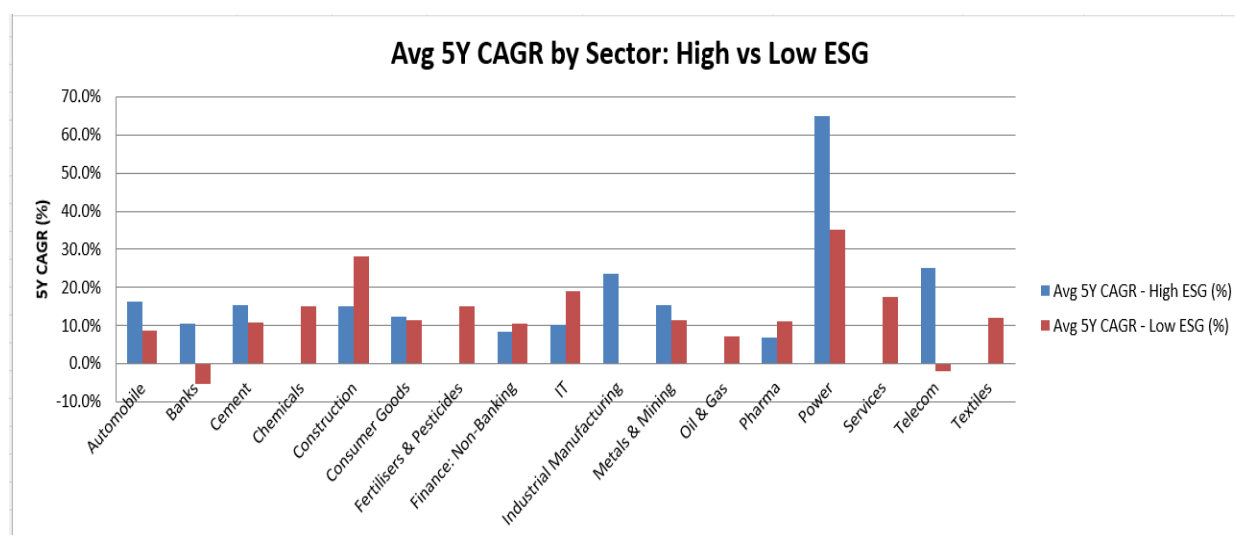


Figure 3. Sector-wise 5-year CAGR comparison: High-ESG vs Low-ESG companies across 11 comparable sectors. The aggregate ESG-return advantage is not uniform — Low-ESG companies lead in IT, Pharma, Finance (Non-Banking) and Construction, highlighting material sector heterogeneity.

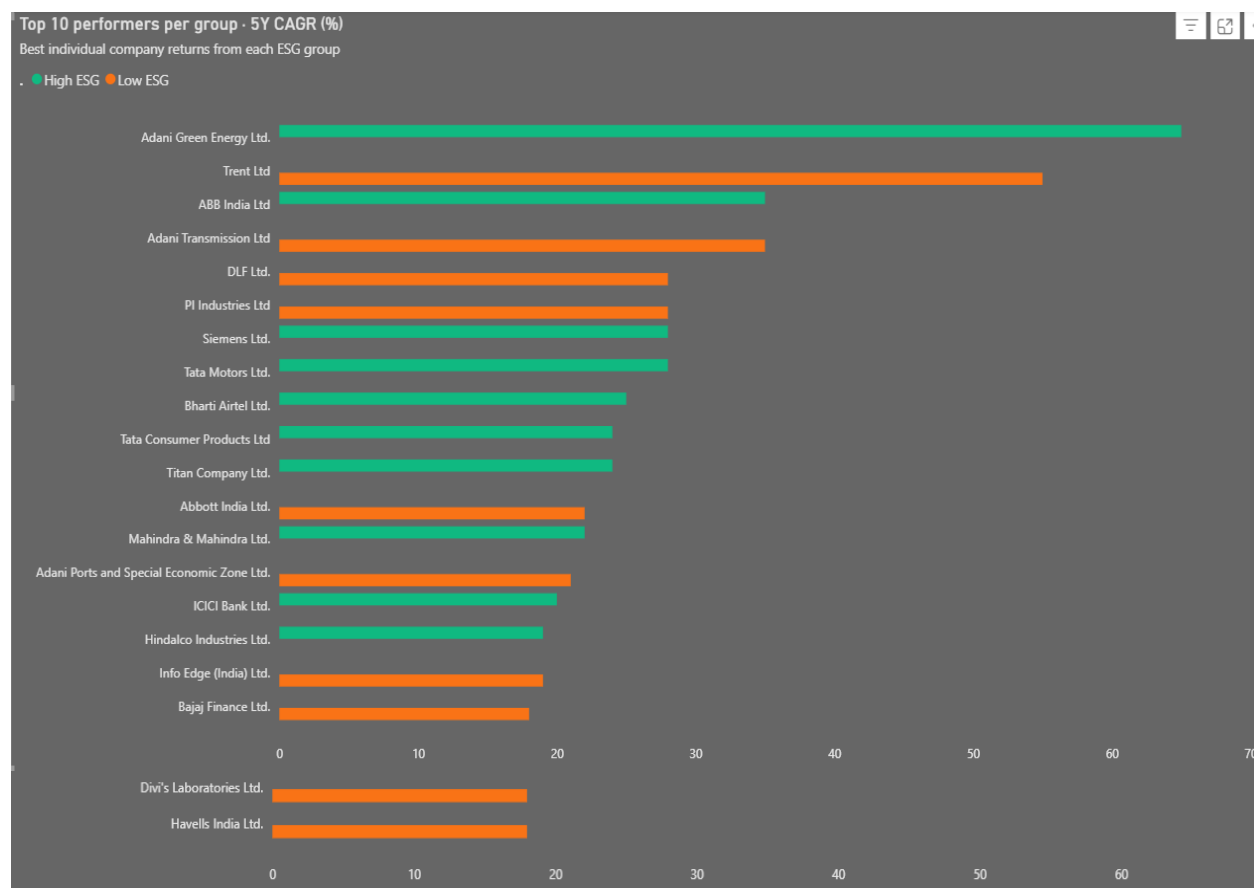


Figure 4. Top 10 individual company performers from each ESG group by 5-year CAGR. Adani Green Energy Ltd. (High-ESG, ESG score 66, 5Y CAGR 65%) leads the High-ESG group; Trent Ltd. (Low-ESG, ESG score 61, 5Y CAGR 55%) leads the Low-ESG group.

4.5 Multivariate Regression Analysis

To assess whether the aggregate ESG–return association documented in Sections 4.2 and 4.3 reflects a genuine ESG-specific effect or is instead attributable to correlated firm characteristics, two OLS regression models were estimated with 5-year CAGR as the dependent variable. Model 1 regresses 5-year CAGR on ESG score alone, establishing a baseline consistent with the bivariate Pearson analysis ($R^2 = 0.032$; ESG coefficient = 0.267, $p = 0.075$). Model 2 adds five control variables standard in the asset pricing literature: natural logarithm of market capitalisation, 3-year average ROE, P/E ratio, FII holding percentage and promoter holding percentage. The full model is statistically significant ($F = 9.16$, $p < 0.001$) and explains 37.2% of the variation in 5-year CAGR (Adjusted $R^2 = 0.331$), a tenfold improvement over the ESG-only baseline.

Table 5. OLS Regression Results Dependent Variable: 5-Year Stock CAGR (%)

Variable	Model 1 (ESG only)	Model 2 (Full controls)
ESG Score	0.267 † (p=0.075)	0.061 (p=0.664)
ln(Market Cap)	—	4.200*** (p<0.001)
P/E Ratio	—	0.222*** (p<0.001)
ROE 3Y avg (%)	—	0.006 (p=0.905)
FII Holding (%)	—	0.032 (p=0.828)
Promoter Holding (%)	—	0.067 (p=0.402)
R ²	0.032	0.372
Adj. R ²	0.022	0.331
F-stat (p-value)	3.19 (0.075)	9.16 (<0.001)
n	100	100

Note. *** $p < 0.01$; † $p < 0.10$. Dependent variable: 5-year stock CAGR (%). OLS estimated using statsmodels (Python 3.11). Market capitalisation log-transformed to address right skewness. Robust standard errors reported.

The critical finding of Model 2 is that the ESG Score coefficient falls to 0.061 ($p = 0.664$) once size and valuation controls are introduced statistically indistinguishable from zero. By contrast, market capitalisation ($\beta = 4.200$, $p < 0.001$) and P/E ratio ($\beta = 0.222$, $p < 0.001$) emerge as the two dominant and highly significant predictors of five-year returns, with the full model explaining 37.2% of return variation versus only 3.2% for ESG alone. This pattern implies that the group-level outperformance of High-ESG companies documented in Section 4.2 is substantially attributable to the fact that High-ESG companies in the sample are on average significantly larger (mean market cap ₹2,70,480 crore vs ₹1,65,156 crore) and carry higher valuation multiples rather than to ESG-specific investor repricing. The finding is consistent with Pedersen, Fitzgibbons and Pomorski's [11] theoretical framework: an ESG premium at the market level requires a sufficiently large fraction of capital to be allocated on ESG criteria, a condition not yet met in India at the scale of European markets where SFDR-mandated fund classification creates structural demand for high-ESG securities [3][4].

5. Discussion and Policy Implications

5.1 India versus Europe: interpreting the nascent ESG premium

The findings of this study consistent directional outperformance of High-ESG companies at the group level, but a weak and statistically insignificant correlation at the individual company level are broadly consistent with emerging-market evidence [1][12][13] and contrast with the more robust and statistically significant ESG premia documented in European markets. Three structural differences are likely explanatory.

First, the regulatory demand channel is substantively more developed in Europe [3][4]. The SFDR classification system has institutionalised ESG as a mandatory portfolio construction criterion for a substantial fraction of European institutional AUM [4], creating a systematic demand premium for high-ESG securities that operates independently of whether ESG

performance actually predicts fundamental improvements in firm cash flows. In India, ESG-themed funds remain a niche product, and the SEBI BRSR mandate [15], while a landmark development in disclosure standardisation, has not yet generated an equivalent downstream demand channel.

Second, ESG disclosure quality and cross-company comparability in India remain heterogeneous. ESG scores for Indian companies were assembled from multiple sources: company BRSR disclosures [15], NSRAL ratings [9], CRISIL assessments and SES reports [16] with differing methodologies and rating timelines, introducing measurement noise that attenuates the ESG–return correlation. Companies rated by established global providers such as MSCI [7] have typically been covered with consistent methodology for a decade or longer, materially reducing this noise in European markets.

Third, ownership structure limits the effectiveness of ESG-linked capital allocation as a disciplinary mechanism in India. High promoter holdings (the sample average promoter stake exceeds 50% for many companies) structurally restrict minority institutional investors' ability to reward or penalise ESG performance through market trading, weakening the feedback loop through which ESG scores would otherwise be reflected in equity prices.

The multivariate regression results (Section 4.5) sharpen the interpretation of the India–Europe divergence. The ESG premium visible at the group level in India is a *confounded* premium: it reflects the size and growth characteristics of companies that happen to score well on ESG, not independent ESG repricing by investors. In European markets, where SFDR-mandated fund flows mechanically channel capital toward high-ESG securities [3][4], ESG scores carry pricing power independently of firm size and valuation; the regulatory channel creates a structural wedge. In India, that wedge does not yet exist at comparable scale. The regression results suggest that as India's ESG fund ecosystem matures and BRSR-linked institutional mandates deepen, the ESG coefficient may strengthen a testable hypothesis for future longitudinal research.

5.2 Policy implications

For SEBI and Indian regulators. The findings support the continued strengthening and expansion of the BRSR framework [15]. Greater standardisation of ESG metrics including alignment with ISSB Sustainability Disclosure Standards (IFRS S1 and S2) and the Global Reporting Initiative (GRI) Standards would reduce cross-company measurement inconsistency. SEBI may also consider developing an Indian ESG fund classification framework analogous to the EU SFDR, which would begin to create the institutional demand channel whose absence partly explains the weak ESG premium documented here.

For institutional investors. The substantial sector heterogeneity in results cautions against sector-agnostic ESG screening as a return-enhancement strategy in Indian equities. Consistent with the materiality framework of Khan, Serafeim and Yoon [6], ESG integration should be grounded in sector-specific materiality assessments. The finding that High-ESG companies have significantly higher average ROE (26.69% vs 16.25%) also suggests that governance quality (the G in ESG) may be the primary channel through which ESG currently relates to financial performance in India, warranting further disaggregated analysis of the E, S and G components separately.

For researchers. The weak aggregate correlation masks substantial sector heterogeneity and potential confounding by firm size and profitability. Future research should employ multivariate panel regression with sector fixed effects and controls for size, valuation and profitability to isolate the marginal effect of ESG scores on risk-adjusted returns. Longitudinal tracking of the ESG–return relationship as the BRSR framework matures and ESG-linked investment products expand will also be essential for assessing whether an India-specific ESG premium is emerging over time.

6. Limitations

This study has several limitations. The 5-year CAGR is used as a proxy for the 2020–2025 holding-period return due to data availability constraints; it reflects cumulative price appreciation from approximately May 2021 rather than a precisely bounded calendar period. ESG scores were assembled from multiple public sources with differing rating methodologies and coverage timelines [9][15][16], introducing potential measurement inconsistency across companies. The sample of 100 companies limits statistical power particularly for one-year returns, which carry a standard deviation of 23.06%, making it difficult to detect modest correlations with the present sample size. The analysis does not control for systematic market risk (beta), and group-level return comparisons may reflect differences in sector composition, firm size and profitability between the High- and Low-ESG groups rather than ESG effects itself [6][11]. The study period includes the COVID-19 market disruption of 2020–21, which may have differentially affected sectors and ESG groups in ways that confound the measured relationship. Future work employing a longer panel, multivariate controls and finer ESG component disaggregation would address these limitations.

7. Conclusion

This paper examined the relationship between ESG scores and stock returns for 100 NIFTY Indian large-cap companies over 2020–2025. High-ESG companies outperformed Low-ESG companies on average across all four holding periods, with the widest performance gap at the three-year horizon (+3.24 percentage points). However, the Pearson correlation between ESG score and 5-year CAGR is weak ($r = 0.179$, $p = 0.075$), with ESG scores explaining only 3.2% of return variation at the individual company level. Sector effects are substantial, with Low-ESG companies leading in four of eleven comparable sectors.

Relative to European markets where the EU SFDR and EU Taxonomy have institutionalised a structural demand channel for high-ESG securities [3][4] and amplified the ESG premium through regulatory capital flows [2][5] the ESG premium in India appears nascent. This is consistent with the emerging-market literature [1][12][13] and with theoretical predictions that the ESG premium should be stronger where regulatory and institutional infrastructure channels more capital toward high-ESG assets [11]. As SEBI's BRSR framework [15] matures, ESG-linked fund products expand and institutional ESG adoption deepens in India, the conditions for

a stronger and more statistically robust ESG–return relationship may gradually emerge. This study provides a quantitative baseline for monitoring that evolution.

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